

Filippo Ficarra

☎ +41 76 231 56 90 / +39 342 64 75 035 ✉ filippoficarra.careers@gmail.com
🌐 <https://filippoficarra.github.io> [in linkedin.com/in/filippo-ficarra](https://www.linkedin.com/in/filippo-ficarra)
🐙 github.com/FilippoFicarra 📅 04.06.2000 📍 Zürich 📄 Permit B

EXPERIENCE

IBM Research Zürich

Jan 2025 - Jul 2025

Research Scientist - Intern

- **Structure Elucidation from “Mixture IR Spectra”** | 📄 | **Paper**
 - * Trained, in collaboration with chemists, a Transformer S2S model to generate the **structure of the target molecules** from *mixture IR spectra* and the molecular formula of the target compound.
 - * Aligned encoder representations with target molecule spectra to boost accuracy, achieving **~34% Top-1** (SOTA, outperforming human experts).
- **IBM Granite for Single Atom Catalysts Synthesis** | *DeepSpeed ZeRO-3, LangChain*
 - * Built a domain-specific **knowledge extraction** system, parsing semi-structured SAC literature via LLM agents (LangChain).
 - * Instruction Tuning of **IBM Granite 3.2 8B** with different input feature for SAC synthesis, multitask using different amount of input features and with n-shot for **RAG** enhancement.
- **Large Scale LLM Training** | *DeepSpeed ZeRO-3*
 - * Training setup in multi-node distributed training with DeepSpeed ZeRO-3. Trained 8B to 405B LLMs on Nvidia GH200 in a Slurm cluster on 2-2048 GPUs.

ETH Zürich

Sep 2023 - Sep 2024

Student Teaching Assistant

- **Large Language Models** | 📄 “Prompting and calibration” tutorial | (Mar 2024 - Sep 2024)
 - * Teaching exercise sessions. Wrote, supervised and graded the final exam.
- **Information Security Lab** | 📄 “Tamarin” tutorial | (Sep 2023 - Jan 2024)
 - * Supervised students during Lab sessions. Tested assignments in advance.

ORBYTA TECH

Mar 2022 - June 2022

Backend Engineer - Intern | *C#, ASP.NET, Entity Framework, MySQL*

- Designing and optimizing REST API for a *Business-to-Business* clothing company. Developed the backend in C# with ABP Framework and MySQL database.

JEToP - Student Association

Nov 2021 - Aug 2022

Mobile developer | *Typescript, React Native*

- Front-end development for a podcast application, multimedia player, routing and pages.

EDUCATION

ETH Zürich

Sep 2022 - Nov 2024

Master in Computer Science | *GPA: 5.30/6.00*

- **Major in Machine Intelligence.**
Relevant Courses: NLP, **Large Language Models**, **Computer Vision**, Probabilistic AI.
- **Minor in Information Security.**
Relevant Courses: **Applied Cryptography**, Information Security Lab, Security of Wireless Networks, Network Security.

Politecnico di Torino

Sep 2019 - Jul 2022

Bachelor in Computer Engineering | *GPA: 110/110 cum laude*

- Studies fully funded by **Merit-based Scholarship**: “Percorso Giovani Talenti”.

RESEARCH

ETH Zürich

Apr 2024 - Oct 2024

Master Thesis - Rycolab | Supervisors: Ryan Cotterell, Alex Warstadt | 

- **Title:** A Distributional Perspective on Word Learning in Neural Language Models.
- Research accepted as a long paper at **NAACL 2025** main conference.
- Developed innovative metrics based on LLMs surprisal for quantifying **LLM word learning** trajectories and comparing to **human language acquisition** patterns.
- Trained GPT-2 models from scratch for different amount of text and different linguistic source.
- Findings: LLMs learn differently from humans, relying more on word's frequency in the training set.

ETH Zürich

Sep 2023 - Mar 2024

Semester Project - Rycolab | Supervisors: Ryan Cotterell, Alex Warstadt

- **Title:** How multimodality influences word acquisition: do images help language models mimic human behavior?
- Analyzed Age of Acquisition in FLAVA models trained with varying vision-to-text ratios.
- Findings: Higher visual input negatively impacts word learning efficiency.

ETH Zürich

Dec 2023 - June 2024

Swiss AI initiative – Rycolab, SwissAI, CSCS cluster

- Contributed to the training data team for a large-scale academic multimodal Vision-Video-LM.
- Collaborated on the model architecture team.

PUBLICATIONS

Filippo Ficara, Ryan Cotterell, and Alex Warstadt. 2025. [A Distributional Perspective on Word Learning in Neural Language Models](#). In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 11184–11207, Albuquerque, New Mexico.

SELECTED PROJECTS

Probabilistic AI |

- Gaussian Processes: Pollution prediction with GP regression, Bayesian optimization via LCB acquisition.
- Reinforcement Learning: Actor-Critic algorithm for pendulum swing training.

Computer Vision |

- Object detection and tracking: BoW vs. CNN classifier, condensation algorithm for dynamic tracking.
- 3D reconstruction: Structure from Motion using Essential matrix estimation and point triangulation.

Sentiment analysis on Tweets | | *TPUs, Google Cloud*

- Ensemble of finetuned **RoBERTa** with sparsity introduced by sparsemax, BERTweet and CLIP text features. Our model achieved **92.62%** of accuracy on the classification task (POS/NEG).

Large Language Models |

- Assignment 1: Theoretical aspects of Language modeling. Special focus on RNNs and Transformers.
- Assignment 2: LLMs finetuning approaches, effect of prompting for complex tasks and **attacks against watermarking** and “secret” text embeddings.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C#, Java, C, SQL, Bash.

Technologies: Docker, Git, IBM Blue Vela, IBM Cloud, LaTeX, Linux, PyTorch.

Interests: Large Language Models, AI and LLM Vulnerabilities, LLM Hallucinations, Cryptography, ML optimisation.